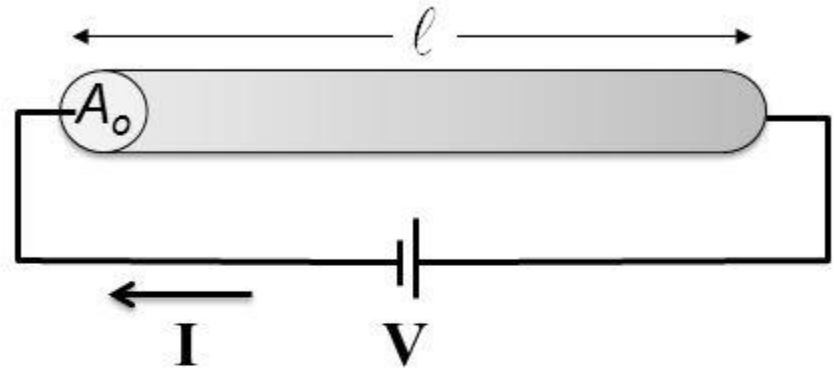
Drift Current
(Arus Hanyut)

Macroscopic Ohm's Law

$$V = IR$$ Ohm's Law

V: voltage (volts = joule/coulomb) **V**
I: current (ampere = coulomb/sec) **A**
R: resistance (ohm = volt/amp) **Ω**



$$R = \rho \frac{\ell}{A_o}$$

$$\rho = \frac{VA_o}{I\ell} = \frac{E}{J}$$

electric field

current density (amp/m^2)

$$\sigma = \frac{1}{\rho}$$

$$J = \sigma E$$ Ohm's Law

*** Don't confuse A for "ampere" with A_o for "cross-sectional area" ***

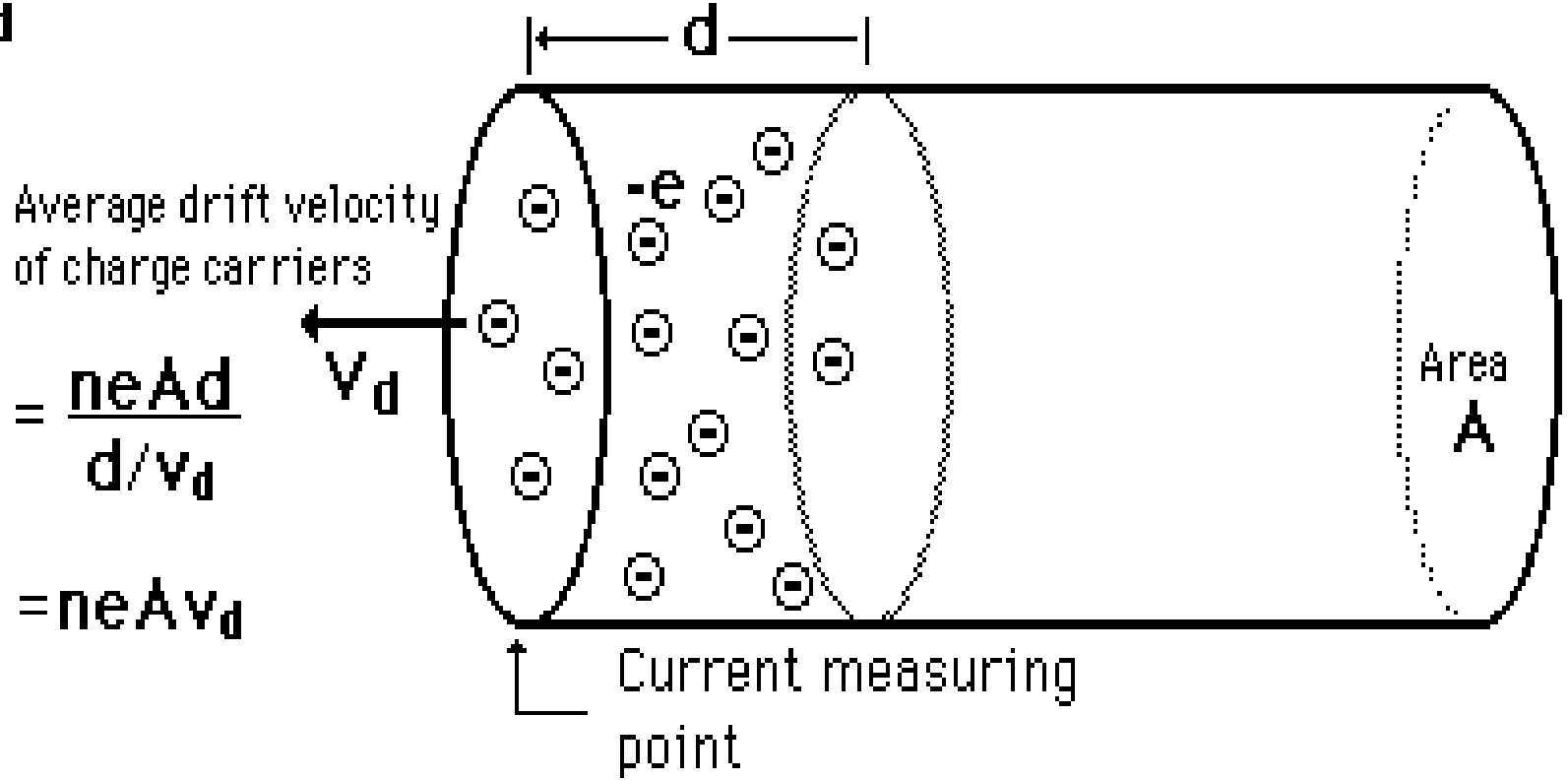
n = number of charges e per unit volume

$Q = neAd$ = total mobile charge in length d of the conductor

$t = \frac{d}{v_d}$ = time for this charge to sweep past the current measuring point

$$I = \frac{Q}{t} = \frac{neAd}{d/v_d}$$

$$I = neAv_d$$



❖ Carrier Drift

- Electron and holes will move under the influence of an applied electric field since the field exert a force on charge carriers (electrons and holes).

$$F = qE$$

- These movements result a current of I_d;

$$I_d = nqV_dA$$

I_d : drift current

V_d : drift velocity of charge carrier

A : area of the semiconductor

n : number of charge carriers per unit volume

q : charge of the electron

Question What causes charges to move in the wire?

How many charges are available to move?

Example What is the drift velocity for 1 Amp of current flowing through a 14 gauge copper wire of radius 0.815 mm?

Drift velocity $v_d = \frac{I}{nqA}$

$n = \rho \frac{N_o}{\mathrm{M}}$ = 8.4x10^{22} atoms/cm^3

$$v_d = \frac{1}{8.4 \times 10^{22} \bullet 1.6 \times 10^{-19} \bullet \pi(.0815)^2}$$

$$v_d = 3.5 \times 10^{-5} \ m/s$$

I = 1 Amp

q = 1.6x10^{-19} C

A = π(.0815 cm)^2

ρ = 8.9 grams/cm ^{3}

N_o = 6x10^{23} atoms/mole

M = 63.5 grams/mole

The higher the density the smaller the drift velocity

❖ Carrier Mobility , μ

$$V_d = \mu E$$

E : applied field

μ : mobility of charge carrier

$$[\mu] = \left[\frac{cm^2}{V - Sec}\right]$$

μ is a proportionality factor

$$\mu = \left[\frac{V_d}{E}\right]$$

❖ So μ is *a measure how easily charge carriers move* under the influence of an applied field or μ determines *how mobile the charge carriers* are.